

Traditional irrigation is more than a production strategy

R code

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```

# PRELIMINARY FUNCTIONS #####

library(sensobol)
sensobol::load_packages(c("data.table", "ggplot2", "sf", "rnatuarearth",
                          "rnatuarearthdata", "ggrepel", "here", "readxl",
                          "ggwordcloud", "cowplot", "benchmarkme", "treemapify"))

# Source all .R files in the "functions" folder -----

r_functions <- list.files(path = here("functions"),
                          pattern = "\\..R$", full.names = TRUE)
invisible(lapply(r_functions, source))

```

1 Interview locations

```
# INTERVIEW LOCATIONS #####

locations <- data.table(
  place = c("Dúrcal", "Huétor-Tajar", "Murcia", "Ricote", "Sudanell", "Arriate", "Oria"),
  lon   = c(-3.5694, -4.0439, -1.1307, -1.4361, 0.6011, -5.0606, -2.2731),
  lat   = c(37.0022, 37.1964, 37.9922, 38.0714, 41.5261, 36.7914, 37.5225)
)

# SPAIN BOUNDARY #####

spain <- ne_countries(scale = "medium", country = "Spain", returnclass = "sf")
```

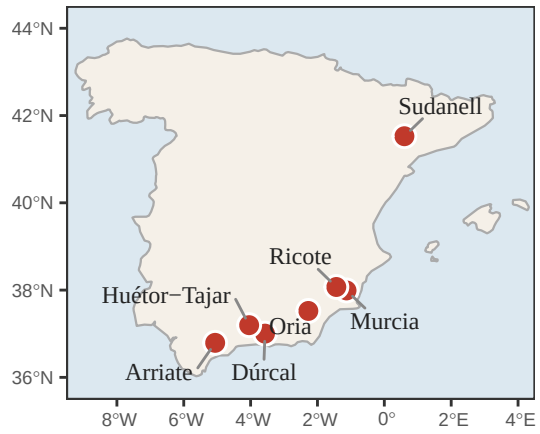
1.1 Plot map

```
# PLOT #####

plot_locations <- ggplot() +
  geom_sf(data = spain, fill = "#f5f0e8", color = "#aaaaaa", linewidth = 0.4) +
  geom_point(data = locations,
    aes(x = lon, y = lat),
    shape = 21, fill = "#c0392b", color = "white",
    size = 3.5, stroke = 0.8) +
  geom_text_repel(data = locations,
    aes(x = lon, y = lat, label = place),
    size = 3.2, family = "serif", color = "#222222",
    box.padding = 0.5, point.padding = 0.3,
    segment.color = "#888888", segment.linewidth = 0.4,
    min.segment.length = 0.2,
    max.overlaps = Inf, seed = 42) +
  coord_sf(xlim = c(-9.5, 4.5), ylim = c(35.5, 44.5), expand = FALSE) +
  labs(title = NULL, x = NULL, y = NULL) +
  theme_AP() +
  theme(
    panel.background = element_rect(fill = "#dce8f0", color = NA),
    panel.grid       = element_line(color = "#cccccc", linewidth = 0.2),
    axis.text        = element_text(size = 8, color = "#555555"),
    plot.margin      = margin(10, 10, 10, 10)
  )
)
```

```
## Warning in geom_text_repel(data = locations, aes(x = lon, y = lat, label =
## place), : Ignoring unknown parameters: `segment.linewidth`
```

```
plot_locations
```



1.2 Load vocabulary data

```
# LOAD VOCABULARY DATA #####

# Irrigators: factors sheet, general list, English translation -----

irrigators <- as.data.table(
  read_xlsx(here("datasets", "input", "carmen_dataset_2.xlsx"), sheet = "factors")
)
irrigators_wc <- irrigators[list == "general" & !is.na(factor_en) & place == "sudanell" ,
  .(n = .N), by = .(word = factor_en)]
irrigators_wc[, group := "Traditional irrigators"]

# Scientists: Sheet2, general list, homogenized concepts -----
# The raw `factor` column has many near-synonyms and a few typos. We map
# them to canonical concepts via a lookup table (raw -> canonical) so the
# wordclouds (and everything downstream) work on homogenized terms.
# See datasets/input/scientists_general_factor_lookup.csv for the mapping.

scientists <- as.data.table(
  read_xlsx(here("datasets", "input", "scientists_vocabulari.xlsx"), sheet = "Sheet2")
)
factor_lookup <- fread(here("datasets", "input",
  "scientists_general_factor_lookup.csv"))
scientists[, factor_canonical := homogenize_factors_fun(factor, factor_lookup)]

scientists_wc <- scientists[list == "general" & !is.na(factor_canonical),
  .(n = .N), by = .(word = factor_canonical)]
scientists_wc[, group := "Scientists"]

# Top 50 per group -----

irrigators_wc <- setorder(irrigators_wc, -n)[1:70]
scientists_wc <- setorder(scientists_wc, -n)[1:70]

# Combine -----

wc_data <- rbind(irrigators_wc, scientists_wc)
```

1.3 Plot wordcloud

```
# WORDCLOUD #####

plot_wordcloud <- ggplot(wc_data, aes(label = word, size = n)) +
  geom_text_wordcloud(seed = 42, color = "black", rm_outside = TRUE) +
  scale_size_area(max_size = 6) +
  facet_grid(. ~ group,
```

```

labeller = as_labeller(c("Traditional irrigators" = "Traditional irrigators",
                          "Scientists" = "Irrigation scientists")) +
theme_AP()

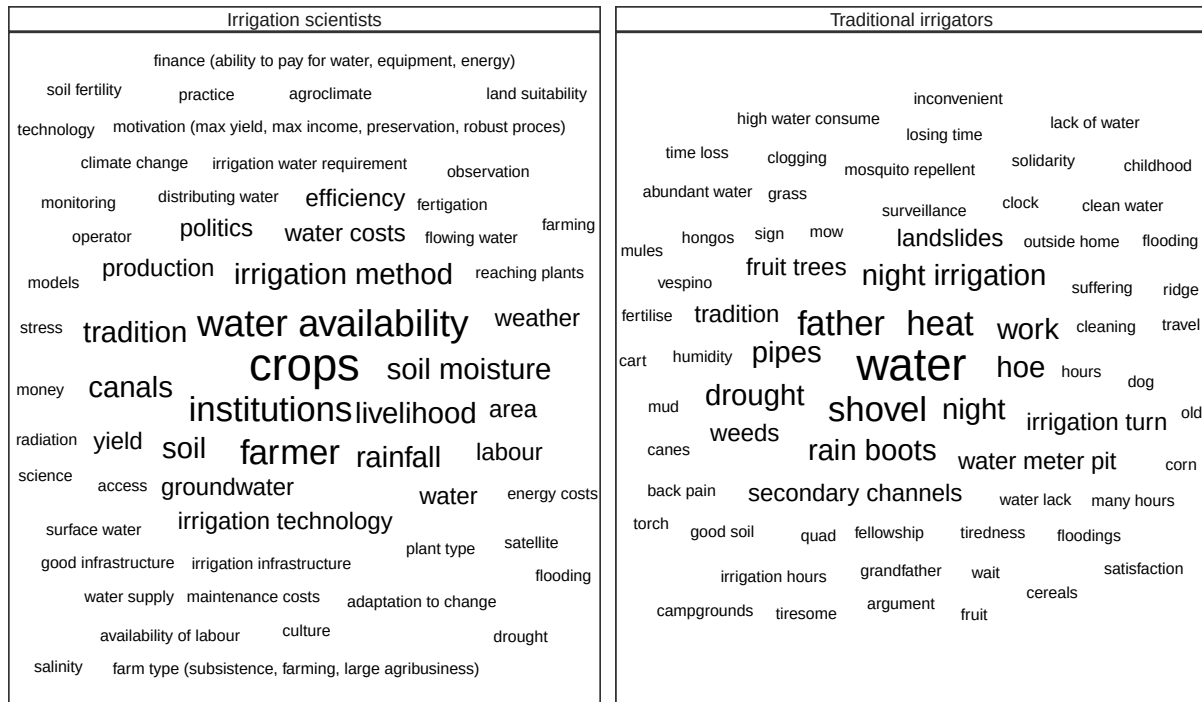
plot_wordcloud

```

```

## Warning in wordcloud_boxes(data_points = points_valid_first, boxes = boxes, :
## Some words could not fit on page. They have been removed.

```



1.4 Pyramid / diverging bar

```

# PLOT PYRAMID / DIVERGING BAR #####

# Words on the y-axis, irrigators counts going left, scientists going right.
# Direct visual comparison: shared vs group-specific terms pop out at once.
wc_wide <- dcast(wc_data, word ~ group, value.var = "n", fill = 0)
setnames(wc_wide, c("Traditional irrigators", "Scientists"),
          c("irrigators", "scientists"))
wc_wide[, total := irrigators + scientists]
wc_pyr <- wc_wide[order(-total)][seq_len(min(.N, 40))]
wc_pyr_long <- melt(wc_pyr, id.vars = "word",
                    measure.vars = c("irrigators", "scientists"),
                    variable.name = "group", value.name = "n")
wc_pyr_long[group == "irrigators", n := -n]

plot_alt_pyramid <- ggplot(wc_pyr_long,
                           aes(x = n,

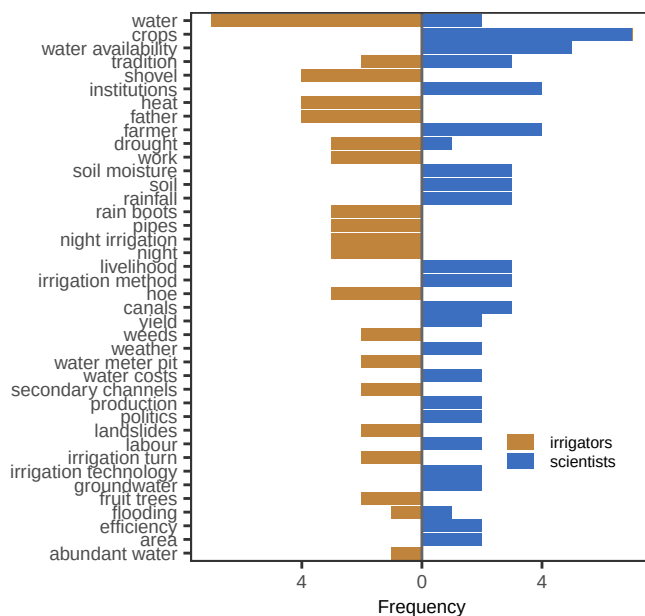
```

```

      y = reorder(word, abs(n), sum),
      fill = group)) +
geom_col() +
geom_vline(xintercept = 0, color = "grey40") +
scale_x_continuous(labels = abs) +
scale_fill_manual(values = c(irrigators = "#c0843d",
                             scientists = "#3d6fc0")) +
labs(x = "Frequency", y = NULL, fill = NULL) +
theme_AP() +
theme(legend.position = c(0.8, 0.2))

```

plot_alt_pyramid



PLOT TREEMAP

```

plot_alt_treemap <- ggplot(wc_data,
                           aes(area = n, label = word, fill = n)) +
  geom_treemap() +
  geom_treemap_text(color = "white", place = "centre",
                    grow = FALSE, reflow = TRUE, min.size = 1) +
  scale_fill_viridis_c(option = "mako", begin = 0.25, end = 0.85,
                       guide = "none") +
  facet_wrap(~ group) +
  theme_AP()

```

plot_alt_treemap

Scientists													
irrigation method	weather	energy costs	satellite	access to support	drought	systems types and behaviours and uniqueness	flexibility vs supply restrictions	system management (mineral and organic)					
		surface water	observation	finance (ability to pay for water, equipment, energy)	soil fertility	systems (new, old, concrete, log, physical)		demand vs supply vs sharing					
		plant type	climate change	turn over (weather, timing, type, agricultural)	planning/scheduling	salinity	water restrictions	local and global knowledge and lack of					
farmer	water	efficiency	radiation	monitoring	availability of labour	land suitability	flooding	technology	practice				
			fertilisation	science	money	water supply	farming	culture	agroclimates				
institutions	area	water availability	irrigation water requirement	maintenance costs	irrigation crop yield (water, equipment, energy, price)	stress	good infrastructure	adaptation to change	models				
			yield	labour	flowing water	distributing water	reaching plants	access	irrigation infrastructure	operator			
crops	soil	groundwater		water costs	irrigation technology	production	politics						
		livelihood	soil moisture	rainfall	tradition	canals							

Traditional irrigators									
night	mow	dog	back pain	losing time	tiresome	inconvenient	childhood		
	cleaning	tiredness	many hours	fertilise	flooding	old	torch		
father	surveillance	outside home	hongos	clogging	high water consume	fruit	lack of water	time loss	
	hours								
shovel		fellowship	canes	solidarity	corn	cart	mules	satisfaction	
	weeds	suffering	grass	clean water	abundant water	ridge	argument	cereals	
heat		water lack	mud	floodings	good soil	travel	wait	irrigation hours	
	tradition	humidity	sign	clock	mosquito repellent	quad	vespino	grandfather	
water	drought		secondary channels	water meter pit	landslides	fruit trees		irrigation turn	
	hoe		rain boots	pipes		night irrigation		work	

1.5 Calculation of Spain's share

We use the ten global irrigated-area datasets compiled by Puy et al. (2026): GAEZ+2015, GIAM, GMIA, GRIPC, LUH2, Meier, MIRCA 2000, MIRCA OS, Nagaraj, and SPAM2010. For each dataset and spatial resolution (0.2°, 0.4°, 1°), we compute Spain's irrigated area (Mha), total European irrigated area (Mha), and Spain's percentage share.

```
# READ IN DATASETS #####

country_level_irrigated_areas <- fread(here("datasets", "input",
                                             "country_level_irrigated_areas.csv"))

# Calculate -----

country_level_irrigated_areas[continent == "Europe",
                              .(spain_mha = sum(mha[country == "Spain"]),
                                europe_mha = sum(mha),
                                spain_share_pct = 100 * sum(mha[country == "Spain"]) /
                                                       sum(mha)),
                              by = .(dataset, resolution)]
```

##	dataset	resolution	spain_mha	europe_mha	spain_share_pct
##	<char>	<char>	<num>	<num>	<num>
## 1:	gaez_v4	0.2deg	3.115963	23.84897	13.065402
## 2:	giam	0.2deg	2.702138	38.65570	6.990269
## 3:	gmia	0.2deg	3.231380	11.95049	27.039725
## 4:	gripc	0.2deg	4.003731	23.03500	17.381072
## 5:	luh2	0.2deg	3.135771	13.78261	22.751647
## 6:	meier	0.2deg	4.223571	26.00185	16.243345
## 7:	mirca2000	0.2deg	3.063020	15.32320	19.989427
## 8:	mirca_os	0.2deg	2.872767	10.92628	26.292260
## 9:	nagaraj	0.2deg	4.651206	32.00352	14.533421
## 10:	spam	0.2deg	2.723286	10.92166	24.934715
## 11:	gaez_v4	0.4deg	3.088649	23.62507	13.073608
## 12:	giam	0.4deg	2.686260	38.53872	6.970287
## 13:	gmia	0.4deg	3.192415	11.81225	27.026305
## 14:	gripc	0.4deg	3.966339	23.03867	17.216008
## 15:	luh2	0.4deg	3.123777	13.73316	22.746234
## 16:	meier	0.4deg	4.177631	25.64511	16.290168
## 17:	mirca2000	0.4deg	3.043205	15.29263	19.899813
## 18:	mirca_os	0.4deg	2.840007	10.87587	26.112921
## 19:	nagaraj	0.4deg	4.601698	31.51188	14.603057
## 20:	spam	0.4deg	2.692517	10.80322	24.923284
## 21:	gaez_v4	1deg	2.936647	23.21744	12.648451
## 22:	giam	1deg	2.558272	38.42329	6.658128
## 23:	gmia	1deg	3.056154	11.54011	26.482897
## 24:	gripc	1deg	3.745961	22.80496	16.426079
## 25:	luh2	1deg	2.948508	13.17117	22.386067

## 26:	meier	1deg	3.928146	25.11399	15.641264
## 27:	mirca2000	1deg	2.887533	14.99172	19.260849
## 28:	mirca_os	1deg	2.712319	10.69960	25.349720
## 29:	nagaraj	1deg	4.272359	30.64785	13.940160
## 30:	spam	1deg	2.559707	10.50107	24.375677
##	dataset	resolution	spain_mha	europa_mha	spain_share_pct
##	<char>	<char>	<num>	<num>	<num>

2 References

Puy, Arnald, Olivia Richards, Seth Nathaniel Linga, Samuel Flinders, Carmen Aguiló-Rivera, and Josh Larsen. 2026. “Code and Datasets of Where Irrigation Exists Is Globally Contested.” *Zenodo*, ahead of print, March. <https://doi.org/10.5281/zenodo.19001232>.

3 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] treemapify_2.5.6      benchmarkme_1.0.8      cowplot_1.2.0.9000
## [4] ggwordcloud_0.6.2     readxl_1.4.5           here_1.0.2
## [7] ggrepel_0.9.6         rnaturalearthdata_1.0.0 rnaturalearth_1.1.0
## [10] sf_1.1-0              ggplot2_4.0.3.9000     data.table_1.18.2.1
## [13] sensobol_1.1.9
##
## loaded via a namespace (and not attached):
## [1] ggfittext_0.10.2      benchmarkmeData_1.0.4  gtable_0.3.6
## [4] xfun_0.55            lattice_0.22-7         vctr_0.7.3
## [7] tools_4.5.2          Rdpack_2.6.6          generics_0.1.4
## [10] parallel_4.5.2       tibble_3.3.1          proxy_0.4-29
## [13] pkgconfig_2.0.3      Matrix_1.7-4          KernSmooth_2.23-26
## [16] RColorBrewer_1.1-3   S7_0.2.2              lifecycle_1.0.5
## [19] stringr_1.6.0        compiler_4.5.2        farver_2.1.2
## [22] tinytex_0.58         codetools_0.2-20      litedown_0.7
## [25] htmltools_0.5.9     class_7.3-23          yaml_2.3.12
## [28] pillar_1.11.1        classInt_0.4-11       iterators_1.0.14
## [31] foreach_1.5.2        commonmark_2.0.0      tidyselect_1.2.1
## [34] digest_0.6.39        stringi_1.8.7         dplyr_1.1.4
## [37] labeling_0.4.3       rprojroot_2.1.1       fastmap_1.2.0
## [40] grid_4.5.2           colorspace_2.1-2      cli_3.6.6
## [43] magrittr_2.0.5       e1071_1.7-17          withr_3.0.2
## [46] scales_1.4.0         rmarkdown_2.30        httr_1.4.8
## [49] otl_0.2.0            cellranger_1.1.0      png_0.1-8
```

```
## [52] evaluate_1.0.5      knitr_1.51           rbibutils_2.4.1
## [55] doParallel_1.0.17    viridisLite_0.4.3    markdown_2.0
## [58] rlang_1.2.0          gridtext_0.1.5       Rcpp_1.1.1-1.1
## [61] glue_1.8.1           DBI_1.3.0            xml2_1.5.2
## [64] rstudioapi_0.17.1    R6_2.6.1             units_1.0-0
```

```
## Return the machine CPU -----
```

```
cat("Machine:    "); print(get_cpu()$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```